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Energetic and Dynamic Analysis of Transport of Na^+ and K^+ through a Cyclic Peptide Nanotube in Water and in Lipid Bilayers

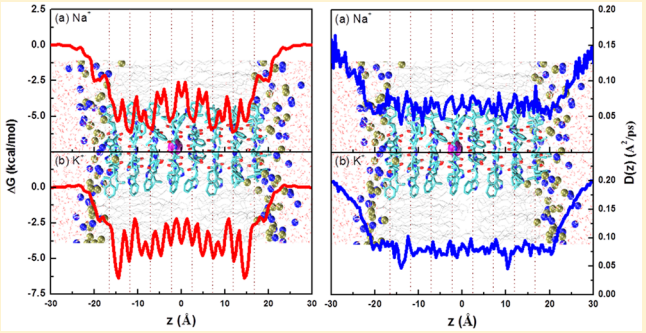
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Supporting Information

ABSTRACT: Potential of mean force (PMF) profiles and position-dependent diffusion coefficients of Na^+ and K^+ are calculated to elucidate the translocation of ions through a cyclic peptide nanotube, composed of 8 × cyclo[-(D-Leu-Trp)$_4$-] rings, in water and in hydrated DMPC bilayers. The PMF profiles and PMF decomposition analysis for the monovalent cations show that favorable interactions of the cations with the CPN as well as the lipid bilayer and dehydration free energy penalties are two major competing factors which determine the free energy surface for ion transport through CPNs both in water and in lipid bilayers, and that the selectivity of CPNs to cations mainly arises from favorable interaction energies of cations with CPNs and lipid bilayers that are more dominant than the dehydration penalties. Calculations of the position-dependent diffusion coefficients and dynamic friction kernels of the cations indicate that the dehydration process along with the molecular rearrangements occurring outside the channel and the coupling of the ion motions with the chain-structured water movements inside the channel lead to a decrease of the diffusion coefficients far away from the channel entrance and also reduced coefficients inside the channel. The PMF and diffusivity profiles for Na^+ and K^+ reveal that the energetics of ion transport through the CPN are governed by global interactions of ions with all the components in the system, while the diffusivity of ions through the channel is mostly determined by local interactions of ions with the confined water molecules inside the channel. Comparison of Na^+ and K^+ ion distributions based on overdamped Brownian dynamics simulations based on the PMF and diffusivity profiles with the corresponding results from molecular dynamics shows good agreement, indicating accuracy of the Bayesian inference method for determining diffusion coefficients in this application. In addition, this work shows that position-dependent diffusion coefficients of ions are required to explain the dynamics and conductance of ions through the CPN properly.



INTRODUCTION

Cyclic peptide nanotubes (CPNs) are a family of synthetic ion channels that are characterized by tubular structures that form through intermolecular hydrogen bonding between cyclic peptide rings.[1−3] Each cyclic peptide ring consists of an even number of D- and L-type amino acid residues which endow CPNs with hydrophobic or hydrophilic properties. Although CPNs have a relatively simple structure compared to natural ion channels, they can function as an ion transporter like natural ion channels and, due to that, they have been extensively studied both experimentally and computationally.[4−17]

From a computational point of view, CPNs provide several advantages as a target system over natural ion channels.[13] First of all, their relatively simple structures enable us to save computational cost, making them a tractable and reliable system for benchmarking theory. Hence, a new theoretical or computational method can be developed by applying the method to a CPN system. Second, the similarity of CPNs to natural ion channels offers an opportunity to better understand the structure and functions of natural ion channels via computational inquiry into CPNs. Third, as a potential antimicrobial peptide (AMP), computational modeling of a large number of CPNs on a membrane surface may provide insight into the mechanisms of action of AMPs.[18] For these reasons, many computational studies have been devoted to examining ion transport through CPNs.[8,11,13,19,20]

The free energy landscape and diffusivity of ions are crucial factors which govern the transport behavior of ions through natural or synthetic ion channels and ion pumps. They can be used to elucidate the ion selectivity of ion channels or to determine ion permeability as can be modeled using Kramers theory.[21−25] They are also required as prerequisite parameters for the calculation of ion conductance across ion channels in

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electrodiffusion modeling such as is described by Poisson−Nernst−Planck (PNP) calculations or in the kinetic lattice grand canonical Monte Carlo (KLGCMC) simulations.[11,26−29] The potential of mean force (PMF) of ions, which is a free energy landscape as a function of ion position, is responsible for the long time evolution of such ions, and can be acquired by time-averaging and integrating the force acting on an ion from all the components in the system.[10,30] On the other hand, the position-dependent diffusion coefficients of ions, which are a dynamic property as a function of ion position, are associated with the fast-varying part of the force acting on an ion, and are governed by the local interaction of an ion with neighboring molecules.[22,31−33] It has been demonstrated that an ion inside ion channels or nanotubes behaves in a different manner than in bulk water due to confinement and, as a result, the free energy and diffusivity of an ion vary with its position.[34−36]

While there are well-established methods such as the umbrella sampling (US), adaptive biasing force (ABF), and steered molecular dynamics (SMD) methods for the calculation of PMF profiles,[10] very few methods for the calculation of position-dependent diffusion coefficients have been developed.[25,35,37] Recently, Comer and co-workers have proposed an efficient and excellent method for position-dependent diffusion coefficients that is based on the Bayesian inference (BI) theorem.[37,38] In that method, trajectories which are obtained from ABF molecular dynamics (MD) simulations are projected into the overdamped Langevin equation, and two parameter sets, one for a PMF profile and the other for a position-dependent diffusion coefficient profile, are optimized simultaneously by employing the Monte Carlo (MC) technique. Comer et al. used the method to obtain position-dependent diffusion coefficients of water through a lipid bilayer membrane. These were found to show qualitative agreement with the results obtained in a previous study.[23,38] They also calculated the diffusivity profiles of water through a CPN embedded in a POPC lipid bilayer.[39]

In this work, we examine the energetic and dynamic behavior of the transport of Na^+ and K^+ through a CPN in water and in lipid bilayers. We calculate the PMF and position-dependent diffusion coefficient profiles for Na^+ and K^+ through a CPN in water and DMPC bilayers using the ABF, TI, and BI/MC methods. The effects of each component in the systems on the energetics and dynamics of ion transport are explained by decomposing the PMF profiles and by calculating dynamic friction kernels of ions which are inversely proportional to diffusion coefficients.[22] Comparisons of ion distributions based on overdamped Brownian dynamics (OBD) calculations with the calculated PMF and diffusion coefficients with the corresponding results of molecular dynamics calculations are used to validate the OBD approach, and to show the importance of using position-dependent diffusion coefficients. Conductances of Na^+ and K^+ through a CPN are calculated, and differences between the calculated conductances and experimentally measured ones are discussed on the basis of the obtained PMF and position-dependent diffusion coefficient profiles.

The paper is organized as follows. In the next section, construction of the simulation systems is introduced and computational methods for the calculation of PMF profiles and position-dependent diffusion coefficients for ions are briefly explained. The calculated results are discussed in the Results and Discussion section, and conclusions are made in the final section.

COMPUTATIONAL METHODS

Description of the Simulations. Two model systems are prepared for the study of ion transport through a CPN which is formed by 8 × cyclo[-(Trp-D-Leu)$_4$-] rings. The first system includes a single CPN, 3355 water molecules, and a single ion such as Na^+ or K^+ in a simulation box with approximately 40.0 Å × 40.0 Å × 70.0 Å dimensions (Figure 1a and c). The second system is composed of a single CPN, 80 DMPC (1,2-dimyristoyl-*sn*-glycero-3-phosphocholine) lipid molecules, and a single ion such as Na^+ or K^+ in a simulation box with approximately 48.0 Å × 48.0 Å × 90.0 Å dimensions (Figure 1b and d). The channel axis of the CPN and the normal to the DMPC lipid bilayers were aligned parallel to the z axis of the systems. All of the simulations were performed using the NpT ensemble and periodic boundary conditions in all directions. The two systems were fully equilibrated at 303 K and 1 bar using a Langevin thermostat with a damping coefficient, γ_L, of 4 ps^{-1} and the Nosé−Hoover Langevin piston method. The diffusion coefficients of ions in the TIP3P water model are generally overestimated compared to the experimental values,[40] but they also depend on the choice of the damping coefficient, γ_L, in the Langevin thermostat. We chose γ_L = 4.0 ps^{-1} because the ABF trajectories of Na^+ and K^+ with that value of γ_L produced diffusion coefficients of 0.141 and 0.203 $Å^2$/ps for the two ions in bulk water, respectively, which are very close to the experimental results.[41] Electrostatic potential energy was treated with the particle mesh Ewald (PME) method. The CHARMM27 force field parameter sets were used for the CPN

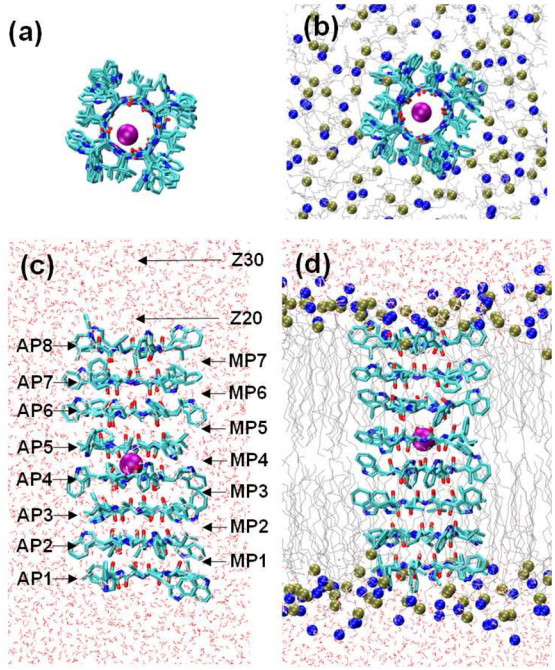


Figure 1. Top views (a and b) and side views (c and d) of a cyclic peptide nanotube (CPN), 8 × cyclo[-(Trp-D-Leu)$_4$-] in water and in DMPC lipid bilayers. Hydrogen atoms in CPNs are removed in all pictures, and water molecules in parts a and b are not shown for clarity. In parts b and d, lipid molecules are highlighted by transparent gray tubes, and P and N atoms in lipid head groups are represented by tan and blue spheres, respectively. In part c, the alpha-plane (AP) regions and midplane (MP) regions are defined in the text following the work in ref 8. Channel entrances are located near z = ±20.0 Å (Z20), and Z30 is defined as the position at z = 30 Å.

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and DMPC, and the TIP3P water model for water.[42,43] The force field parameters for the ions were obtained from Beglov and Roux.[44] All of the energy minimizations and MD simulations were performed using NAMD 2.10,[45] and visualizations were made possible using VMD 1.9.0.[46]

Construction of PMF Profiles. PMF profiles of ions, $\Delta G(z)$, along the channel axis were constructed using the adaptive biasing force (ABF) method. The channel axis was restrained parallel to the z direction via a colvar method. It is known that the channel axis of CPNs in lipid bilayers is not parallel but tilted with respect to the membrane normal,[14] but in this study, the effect of the tilt behavior of CPNs on the PMF and position-dependent diffusion coefficient profiles were not taken into account. The distance between the two adjacent cyclic peptide rings was kept at 4.75 Å in the z direction by restraining the center of mass of the alpha carbons in each ring at ±16.625, ±11.875, ±7.125, and ±2.375 Å with a harmonic spring constant of 25.0 kcal/mol/Å^2. The one-dimensional PMF curve is not well-defined outside the channel because the sampling over all of the xy plane outside of the channel causes the PMF profile to diverge to $-\infty$.[24,35] To avoid the problem due to the integration over all xy extents and to keep the ions within a certain area as they exit the channel, a flat-bottom half-harmonic restraint was applied to the ions in the xy plane as shown below

$$V_{\mathrm{fb}}(x, y) = \begin{cases} 0 & \text{if } x^2 + y^2 \le {R_0}^2 \\ \frac{1}{2} k_{xy}^{\mathrm{ion}} (\sqrt{x^2 + y^2} - R_0)^2 & \text{otherwise} \end{cases} \tag{1}$$

where k_{xy}^{ion} = 10 kcal/mol/Å^2 and R_0 = 2.5 Å for the ions.[47] Instead of the flat-bottom half-harmonic restraint, a conventional harmonic restraint is sometimes applied to ions in the xy plane.[12,13]

The reaction pathway of each ion spanned from −32.0 to 32.0 Å and was divided by 161 bins with a bin width of 0.4 Å. For a PMF profile and a position-dependent diffusion coefficient for each ion under a specific condition, two or three independent simulations were carried out and in each simulation no fewer than 1 000 000 samples were collected per bin. The final symmetrized PMF profiles shown in the next section were constructed by creating duplicate bins on opposite sides of the channel.

In order to examine each contribution from the CPN, water, and DMPC bilayers to the energetics of ion transport, we decomposed the total PMF profile of each ion by using the thermodynamic integration (TI) method, namely, by computing and integrating the mean force acting on the ion from the CPN, water, and DMPC bilayers separately

$$\Delta G_\alpha(z) = G_\alpha(z) - G_\alpha(z_0) = -\int_{z_0}^{z} \langle F_\alpha(z') \rangle_{z'} \, \mathrm{d}z' \tag{2}$$

where $F_\alpha(z)$ is the force acting on the ion at the position of z by a component α. The forces acting on an ion from each component were collected from all of the corresponding ABF MD trajectories.

Calculation of Position-Dependent Diffusion Coefficients. Position-dependent diffusion coefficients of ions along the channel axis (z axis), $D(z)$, were calculated using the Bayesian inference (BI) method proposed by Comer et al.[37,39] Since the method was described in detail in ref 37, we only briefly address the method. The BI method uses the assumption that the ion in an ABF simulation undergoes a free diffusional motion under a given biasing force and that the motion can be completely determined by a PMF and a position-dependent diffusion coefficient. The Bayesian inference method is based on the Bayes' theorem

$$P(\{a_i\}, \{b_i\} | \{z(t)\}) \propto P_{\mathrm{L}}(\{z(t)\} | \{a_i\}, \{b_i\}) \times P_{\mathrm{prior}}(\{a_i\}, \{b_i\}) \tag{3}$$

where $\{z(t)\}$ represents the set of z coordinates of an ion obtained from an ABF MD simulation and $\{a_i\}$ and $\{b_i\}$ are parameter sets needed to construct $D(z)$ and $\Delta A(z)$. Equation 3 shows that the suitable $\{a_i\}$ and $\{b_i\}$, which fit well for $D(z)$ and $\Delta A(z)$, respectively, can be obtained by choosing the best likelihood function, $P_{\mathrm{L}}(\{z(t)\}|\{a_i\}, \{b_i\})$, and the best prior probability, $P_{\mathrm{prior}}(\{a_i\}, \{b_i\})$. A Monte Carlo (MC) technique is employed to find the set $\{a_i\}$ and $\{b_i\}$ which produces the likelihood function and prior probability with the highest values.[37] The likelihood function, $P_{\mathrm{L}}(\{z(t)\}|\{a_i\}, \{b_i\})$, is defined as

$$P_{\mathrm{L}}(\{z(t)\}|\{a_i\}, \{b_i\} = \prod_{n=0}^{N} P(z(t + \Delta t)|z(t), \{a_i\}, \{b_i\}) \tag{4}$$

where Δt, which is called the observation time or lag time, is not the simulation time step but the time step in the BI method, which can be adjusted, N is the total number of the BI steps, and $t = n\Delta t$. In eq 4, $P(z(t + \Delta t)|z(t), \{a_i\}, \{b_i\})$ is given as

$$P(z(t + \Delta t)|z(t), \{a_i\}, \{b_i\}) = \frac{1}{\sqrt{4\pi D(z(t))\Delta t}} \exp\left[-\frac{\{z(t + \Delta t) - z_{\mathrm{OLE}}(t + \Delta t)\}^2}{4D(z(t))\Delta t} \right] \tag{5}$$

where $z(t + \Delta t)$ is the z coordinate of the ion at time $t + \Delta t$ obtained from a single ABF trajectory and $z_{\mathrm{OLE}}(t + \Delta t)$ is the z coordinate of the ion at $t + \Delta t$ that is estimated using the overdamped Langevin equation (OLE), where $z_{\mathrm{OLE}}(t + \Delta t)$ is defined as

$$z_{\mathrm{OLE}}(t + \Delta t) = z(t) + \frac{D(z(t))}{k_{\mathrm{B}}T} \left[f_{\mathrm{bias}}(z(t)) - \left(\frac{\mathrm{d}\Delta G(z)}{\mathrm{d}z} \right)_{z=z(t)} \right] \Delta t + \nabla D_k(z(t)) \Delta t \tag{6}$$

Here $f_{\mathrm{bias}}(z)$ is the biasing force that can be obtained from an ABF MD simulation, and $D(z)$ and $\Delta G(z)$ are position-dependent diffusion coefficient and PMF profiles that are constructed using $\{a_i\}$ and $\{b_i\}$ at the kth MC cycle. As Comer et al. proposed, we used $P_{\mathrm{prior}}(\{a_i\}, \{b_i\})$ given by

$$P_{\mathrm{prior}}(\{a_i\}, \{b_i\}) = P_{\mathrm{scale}}(\{a_i\}) P_{\mathrm{smooth}}(\{a_i\}, \{b_i\}) \tag{7}$$

where $P_{\mathrm{scale}}(\{a_i\}) = \prod_i 1/a_i$ and $P_{\mathrm{smooth}}(\{a_i\}, \{b_i\})$ is defined as

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$$P_{\text{smooth}}(\{a_i\}, \{b_i\}) = \sum_{i=2}^{n} \exp\left[-\frac{(a_i - a_{i-1})^2}{2\epsilon_D^2} - \frac{(b_i - b_{i-1})^2}{2\epsilon_w^2} \right] \quad (8)$$

In our calculations, Δt = 1.2 ps was chosen. It is worth noting that the dependence of the diffusivity of ions on Δt implies that the motion of ions is not Markovian and that a nondiffusive model may be required to describe the dynamics of ions properly.[25,48] The non-Markovian behavior of ions in a confined region such as ion channels is generally observed.[49] Indeed, the calculated profiles for the position-dependent diffusion coefficients of Na^+ and K^+ inside the CPN show a slight dependence on Δt (Figure S1), which indicates that the ion motions inside the CPN are not completely Markovian. The functions $D(z)$ and $\Delta G(z)$ were discretized and interpolated by piecewise cubic polynomials with a bin width of 0.4 Å from −32.0 to 32.0 Å.

RESULTS AND DISCUSSION

Figure 2 represents the symmetrized PMF profiles for a single Na^+ and K^+ through the model CPN in water and in DMPC

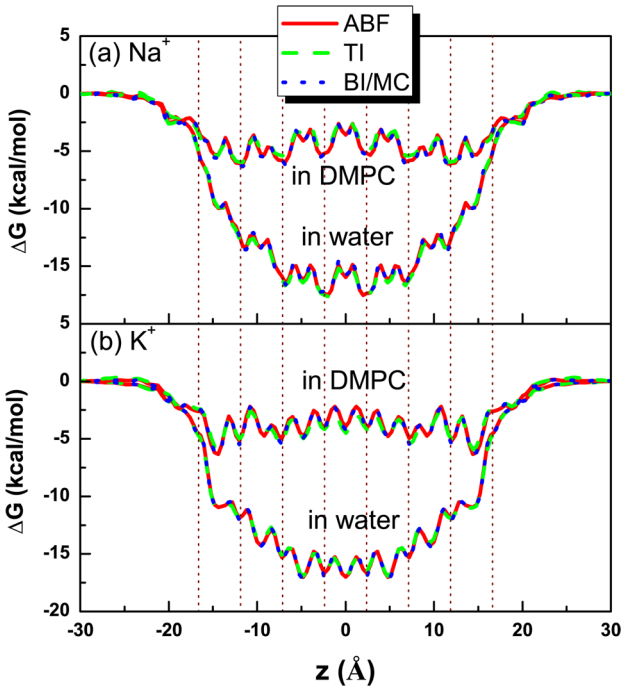


Figure 2. Final symmetrized PMF profiles of (a) Na^+ and (b) K^+ through the CPN in water and in DMPC bilayers from the ABF, TI, and BI/MC methods, respectively. See the text for details. The vertical broken lines indicate the position of each cyclic peptide ring in the z direction.

bilayers that have been constructed using the ABF, TI, and BI/MC methods, respectively. The PMF curves calculated from each method show good agreement with those from the other methods. The PMF profiles for all of the ions feature several local energy barriers and energy wells that indicate weak binding sites; thus, ion motions inside CPNs can be envisioned as hopping from one binding site to another. Previous studies have shown that energy wells in addition to energy barriers can greatly affect ion transport through channels and that the energy well depth and width as well as the number of binding sites play important roles in determining transport rates.[50,51] The PMF curves for both cations indicate that the cations are more stabilized through the CPN in water than in DMPC bilayers. This can be explained by the fact that cations are more stabilized in a high dielectric medium like water than in low dielectric lipid bilayers. The free energy values for K^+ in the middle of the CPN both in water and in DMPC bilayers are slightly lower than those for Na^+, but the differences are minimal. This implies that the higher conductance of K^+ over Na^+ through the CPN in experiments cannot be easily explained by the PMF profiles alone.[2]

The PMF decomposition profiles in Figure 3 demonstrate that there are two competing factors that determine the energetics of the cations passing through the model CPN. One factor is favorable or attractive interactions of cations with CPNs and lipid bilayers, and the other factor is dehydration energy penalties that appear when an ion enters into or stays inside the channel. The favorable interactions of Na^+ and K^+ with the CPN and lipid molecules, which generally lower the free energy, are more dominant than dehydration penalties, making CPNs selective to cations. The two competing factors are greatly enhanced at both channel entrances, but they become moderate as the cations approach the middle of the channel. Note that the rapid lowering of the attractive free energies and the corresponding increase of dehydration penalty energies as a cation approaches the channel entrances ensue from the strongly attractive interactions between cations and the negatively charged dangling carbonyl oxygens located at the outermost peptide rings and the repelled water molecules due to the strong attractive interactions between the cations and carbonyl oxygens.

The dehydration energy penalties are reduced or even disappear in the CPN in water as the cations exist in the middle of the channel. This can be explained by the fact that the dipole moments of water molecules inside the channel reorient in such a way that their electrostatic interactions with cations can stabilize the cations in the middle of the channel.[17]

Position-dependent diffusion coefficients of Na^+ and K^+ for the model CPN in water and in lipid bilayers are presented in Figure 4. To calculate the position-dependent diffusion coefficients of the ions, the BI/MC algorithm combined with the ABF simulation trajectories were used.[37] The position-dependent diffusion coefficients of the ions were also plotted using a fitting function given as

$$D^{\text{fit}}(z) = D_0\left(a_1 + \frac{1.0 - a_1}{1.0 + e^{-(|z|-z_0)/\Delta z}}\right) \quad (9)$$

where D_0 is the bulk diffusion coefficient of the ion and a_1, a_2, z_0, and Δz are fitting parameters whose values are provided in Table 1. The diffusion coefficients of Na^+ inside the channel in Figure 4a are very comparable to those obtained using an overdamped harmonic oscillator framework in a previous MD simulation study.[19]

The calculations show that the position-dependent diffusion coefficients for both ions begin to decrease at more than 10 Å away from the channel entrance and reach their minimum values at the channel entrance.[52] This is related to earlier observations where it was noted that dehydration processes occurring in very narrow ion channels like CPNs can reduce ion mobility.[21] Considering that the onset of favorable interaction energies of the ions with the CPN and lipid bilayers

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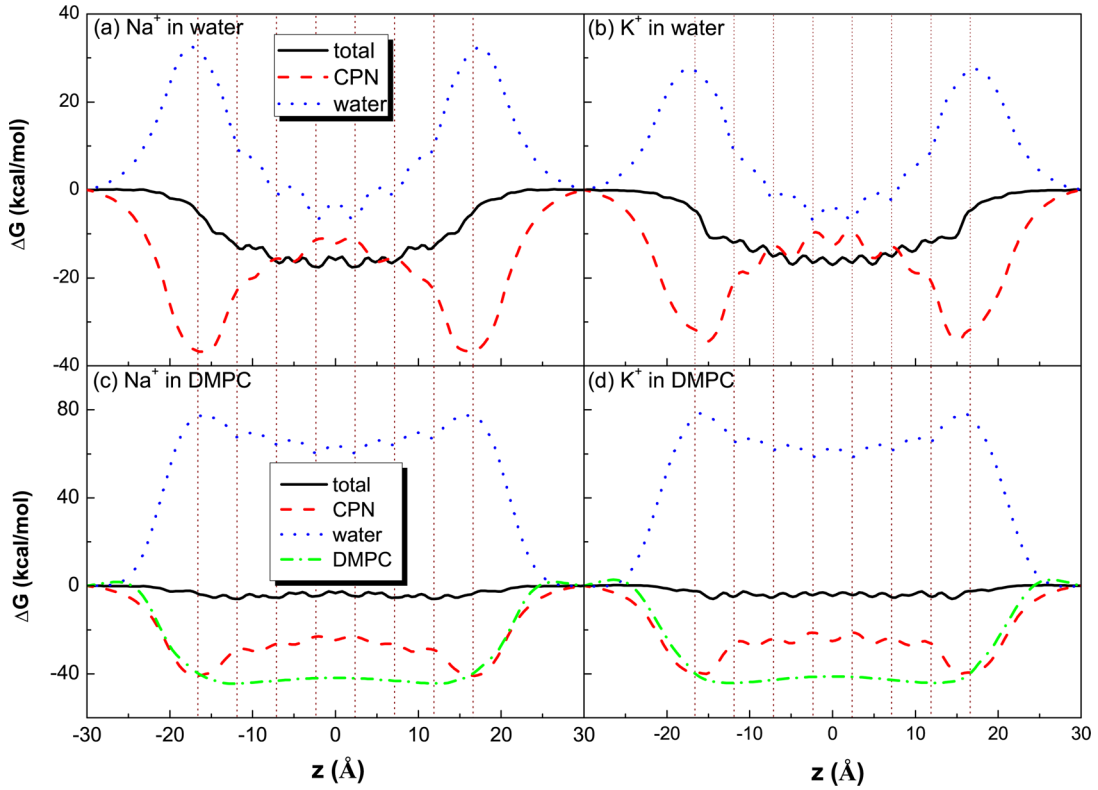


Figure 3. PMF decomposition profiles of Na^+ and K^+ in water (a and b) and in DMPC bilayers (c and d) showing the contributions from the CPN, water, and DMPC lipids to the total PMF profiles. The vertical broken lines are the same as those in Figure 2.

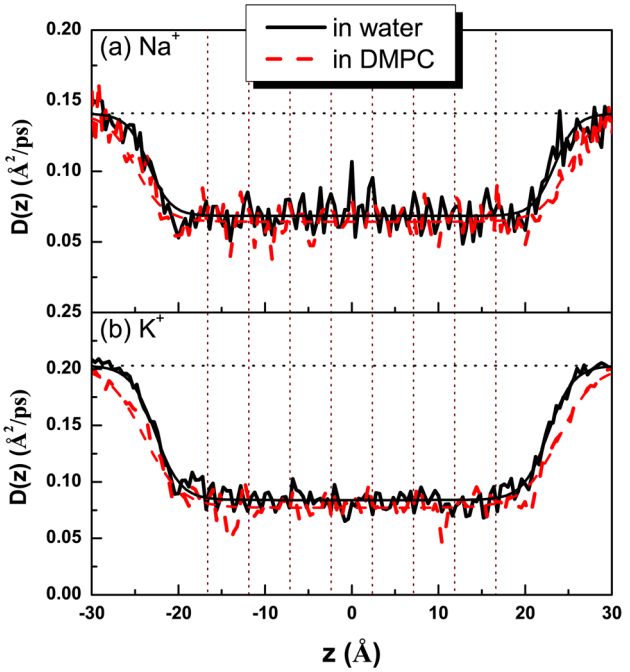


Figure 4. Position-dependent diffusion coefficient profiles of (a) Na^+ and (b) K^+ in water and in DMPC bilayers as a function of axial direction z. The horizontal broken lines indicate the diffusion coefficients of the ions in bulk water that were obtained using the ABF simulations followed by the BI/MC algorithm. The smooth lines are fits to the diffusivity functions that were plotted with eq 9.

and the increase of the dehydration penalty energy for the cations begin more than 10 Å away from the channel entrance in the PMF profiles, we can speculate that the reduction of the

Table 1. Parameters of the Fitting Function for the Position-Dependent Diffusion Coefficients of Na^+ and K^+ in eq 9

	Na^+		K^+	
	in water	in DMPC	in water	in DMPC
a_1	0.485	0.456	0.414	0.380
z_0	23.4	24.9	22.7	24.0
Δz	1.26	1.64	1.44	2.10

position-dependent diffusion coefficients of the ions is associated with the rate of dehydration and molecular rearrangement processes due to the free energy variation. Single-file water movements in a narrow channel can also lower ion mobility because the motions of ions are coupled by the single-file water structure.[17,21] Previous MD simulations have reported that water molecules inside a CPN are not completely in a single-file but possess a unique structure such as 1−2−1−2 water chains.[8,20,53] Thus, the reduction of the position-dependent diffusion coefficients of the ions inside the channel can be attributed to coupling of the motions of the ions to the chain-structured water movements inside the channel. Figure 4 also reveals that, despite a striking difference in the PMF profiles for each ion through the CPN in water and in DMPC lipid bilayers, the diffusion coefficient profiles for each ion are much alike regardless of whether the CPN is in water or in DMPC bilayers. This implies that, unlike the PMF profiles which are generally governed by interactions of ions with global environments, the position-dependent diffusion coefficients of ions are more affected by local interactions of ions with water through confinement.

To assess the contribution of each component to the position-dependent diffusion coefficient of the ions, we calculated the dynamic friction kernel assuming that the

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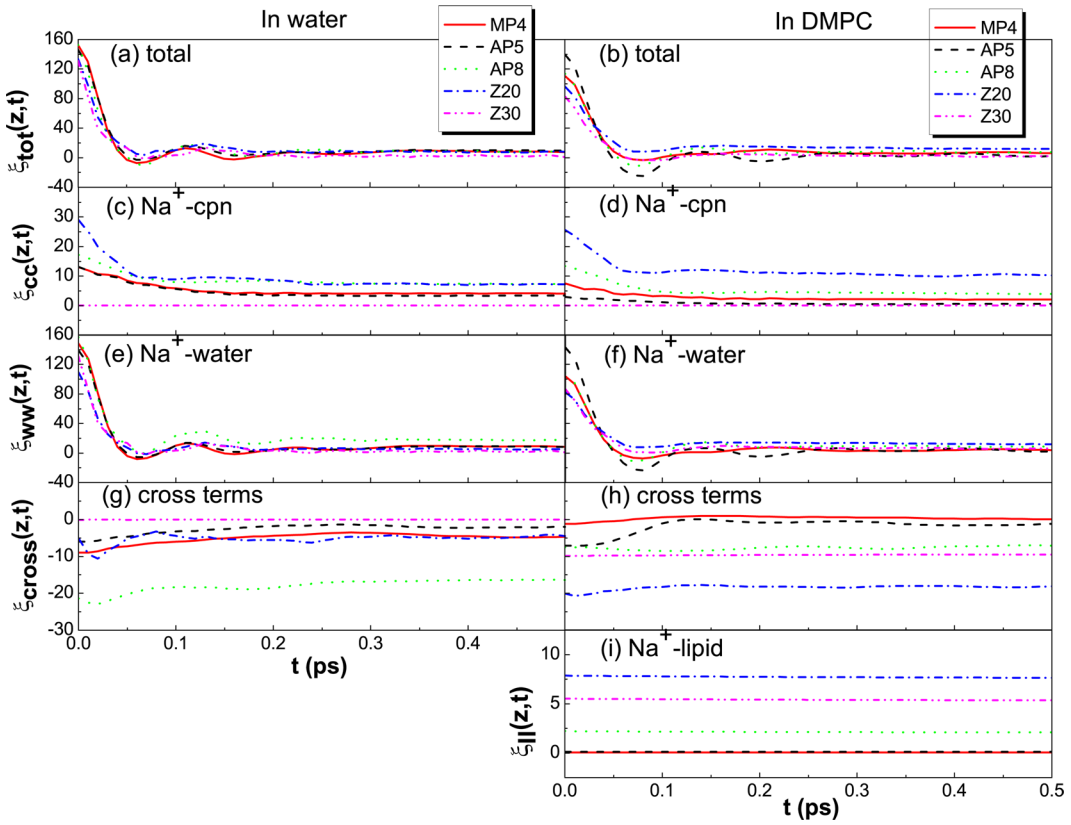


Figure 5. Total and decomposed dynamic friction kernels for Na^+ at a given position inside or outside the CPN in water and in hydrated DMPC bilayers. For the definition of symbols in the legend, see Figure 1.

dynamics of ions inside the channel can be described by the Brownian dynamics method. The position-dependent diffusion coefficient $D(z)$ of an ion is associated with the dynamic friction kernel $\xi(z, t)$ via the following equation:[36]

$$D(z) = \frac{k_B T}{\int_0^\infty \xi(z, t)\, dt} \tag{10}$$

On the other hand, the dynamic friction kernel can be expressed using the autocorrelation function of the random force $\delta F(z)$ acting on an ion[22,36]

$$\xi(z, t) = \frac{\langle \delta F(z, t) \delta F(z, 0) \rangle}{k_B T} \tag{11}$$

where $\delta F(z, t) = F(z, t) - \langle F(z) \rangle$ and $F(z, t)$ is the force acting on the ion at the position of z at time t. $\langle A \rangle$ means the time average of a physical variable A. Because the random force on an ion can be decomposed into the contribution of each component, the dynamic friction kernel can also be decomposed as

$$\begin{aligned} \xi(z, t) &= \frac{\langle (\delta F_c(z, t) + \delta F_w(z, t) + \delta F_l l(z, t))(\delta F_c(z, 0) + \delta F_w(z, 0) + \delta F_l(z, 0)) \rangle}{k_B T} \\ &= \xi_{cc}(z, t) + \xi_{ww}(z, t) + \xi_{ll}(z, t) + \xi_{cross\text{-}terms}(z, t) \end{aligned} \tag{12}$$

where $\xi_{jj}(z, t)$ is defined as

$$\xi_{jj}(z, t) = \frac{\langle \delta F_j(z, t) \delta F_j(z, 0) \rangle}{k_B T} \tag{13}$$

and $\xi_{cross\text{-}terms}(z, t)$ is defined as

$$\begin{aligned} \xi_{cross\text{-}terms}(z, t) = \frac{1}{k_B T} [&\langle \delta F_c(z, t) \delta F_w(z, 0) \rangle \\ &+ \langle \delta F_c(z, t) \delta F_l(z, 0) \rangle + \langle \delta F_w(z, t) \delta F_c(z, 0) \rangle \\ &+ \langle \delta F_w(z, t) \delta F_l(z, 0) \rangle + \langle \delta F_l(z, t) \delta F_c(z, 0) \rangle \\ &+ \langle \delta F_l(z, t) \delta F_w(z, 0) \rangle] \end{aligned} \tag{14}$$

As a result, the contribution of each component to the position-diffusion coefficient can be examined through eqs 10, 13, and 14.

To acquire the dynamic friction kernels, we performed a series of 5 ns MD simulations for each system using the heavy particle approximation. In the approximation, a cation was fixed at a specific position and forces acting on the cation from the total system or from a component of the system were calculated as a function of time.[54,55] Figure 5 reveals that the largest contribution to the total dynamic friction kernels (Figure 5a and b) comes from the water−water dynamic friction kernel (Figure 5e and f), demonstrating that the position-dependent diffusion coefficients of the cations through the channel are mostly affected by local interactions with water molecules

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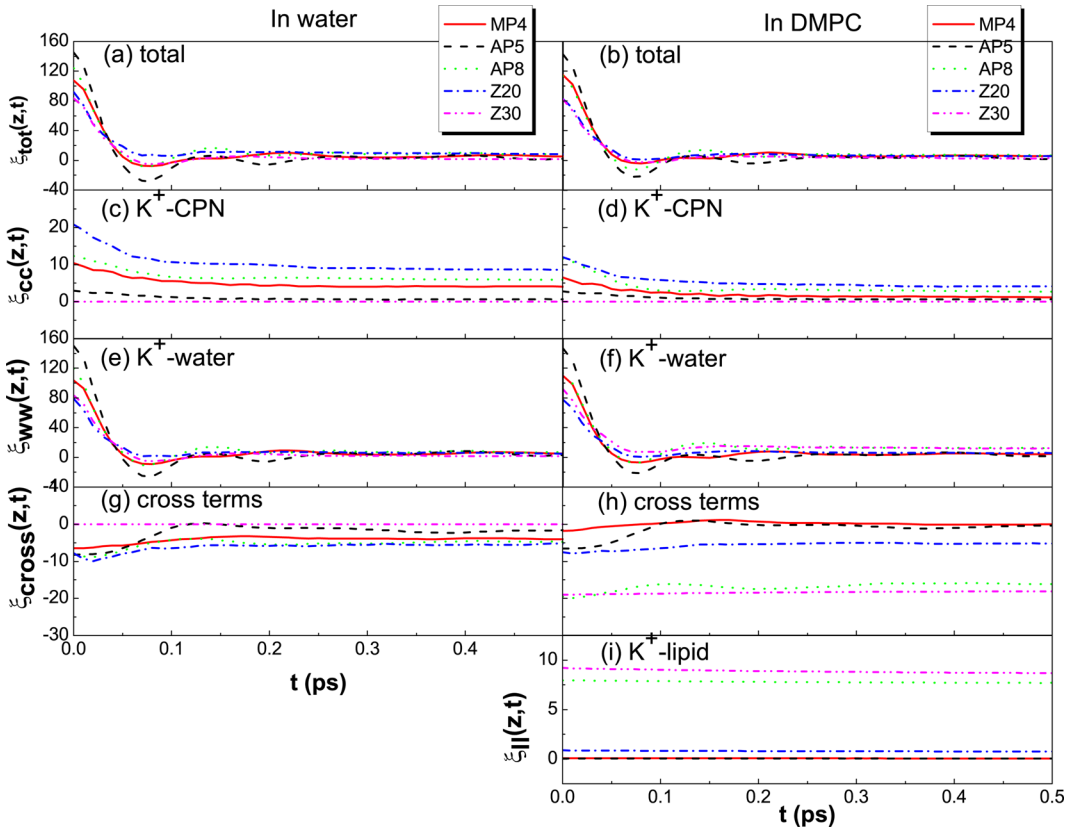


Figure 6. Total and decomposed dynamic friction kernels for K^+ at a given position inside or outside the CPN in water and in hydrated DMPC bilayers. For the definition of symbols in the legend, see Figure 1.

around the cations. In Figure 5a and b, the total dynamic friction kernels for Na^+ inside and outside the CPN in water and in DMPC bilayers are quite similar at the corresponding positions, implying the similarity of the position-dependent diffusion coefficients of Na^+ through the CPN in water and in bilayers as in Figure 4. In Figure 5e and f, the higher values of the CPN−CPN dynamic friction kernels at the Z20 position than at any other positions indicate the importance of dynamic interactions of Na^+ with the dangling carbonyl oxygens at the outermost cyclic peptide rings. The oscillating behavior of the water−water dynamic friction kernels at the AP5 or AP8 position in Figure 5e and f can be viewed as Na^+ at the AP5 or AP8 position interacting with water molecules bouncing at MP4 and MP5 or at MP7 and Z20 positions. Because PMF calculations for the translocation of a single water molecule through the CPN showed free energy wells of 0.4 kcal/mol at the MP3, MP4, MP7, and Z20 locations,[39] we can infer that water molecules at those positions in the presence of neighboring K^+ bounce back and forth in those energy wells, leading to the oscillating behavior of the water−water dynamic friction kernels. Similar behavior in the dynamic friction kernels for K^+ in Figure 6 are observed.

The calculated PMF and position-dependent diffusion coefficient profiles for Na^+ and K^+ can be combined to estimate ion conductances γ through the CPN using the equation[56]

$$\gamma = \frac{I}{V} = \frac{q^2 c_0 (\pi \langle r_s^2 \rangle)}{k_B T R_p} \tag{15}$$

where q, c_0, T, and $\pi\langle r_s^2\rangle$ are the ion charge, bulk ion concentration, temperature, and the effective cross-sectional area inside and outside the channel, respectively. Here the permeability resistance R_p is defined as

$$R_p = \int_{z_i}^{z_f} dz \frac{\exp[\Delta G(z)/k_B T]}{D(z)} \tag{16}$$

where $\Delta G(z_i) = \Delta G(z_f) = 0$ in the bulk and $z_i = -30.0$ Å and $z_f = 30.0$ Å.[57] The parameters that we used in the conductance calculations are provided in Table 2. At first, we took $R_0 = 2.5$ Å

Table 2. Comparisons of Calculated and Measured Conductances of Na^+ and K^+ through a CPN Composed of 8 × cyclo[-(Trp-D-Leu)$_4$-] Rings in a DMPC Bilayer

					γ (pS)	
ion	q (|e| C)	c_0[a] (M)	$\langle r_s^2\rangle$ (Å^2)	R_p (ps/Å)	calc.	exp.[a]
Na^+	+1	0.5	6.25[b]	136 (144)[c]	270 (256)	55
K^+	+1	0.5	6.25	130 (133)	284 (277)	65

[a]Taken from ref 2. [b]Taken from $R_0 = 2.5$ Å of the flat-bottom half-harmonic constraint in eq 1. [c]Calculated with $D^{fit}(z)$ of eq 9 in parentheses.

used in the flat-bottom half-harmonic restraint as the radius r_s of the effective cross-sectional area. We evaluated the conductance of Na^+ and K^+ through the CPN in the hydrated DMPC bilayers using the obtained PMF and position-dependent diffusion coefficient profiles for the ions and compared the calculated conductances with the experimentally measured ones in Table 2.[2] Here we assume that the PMF and position-dependent diffusion coefficients of ions calculated at

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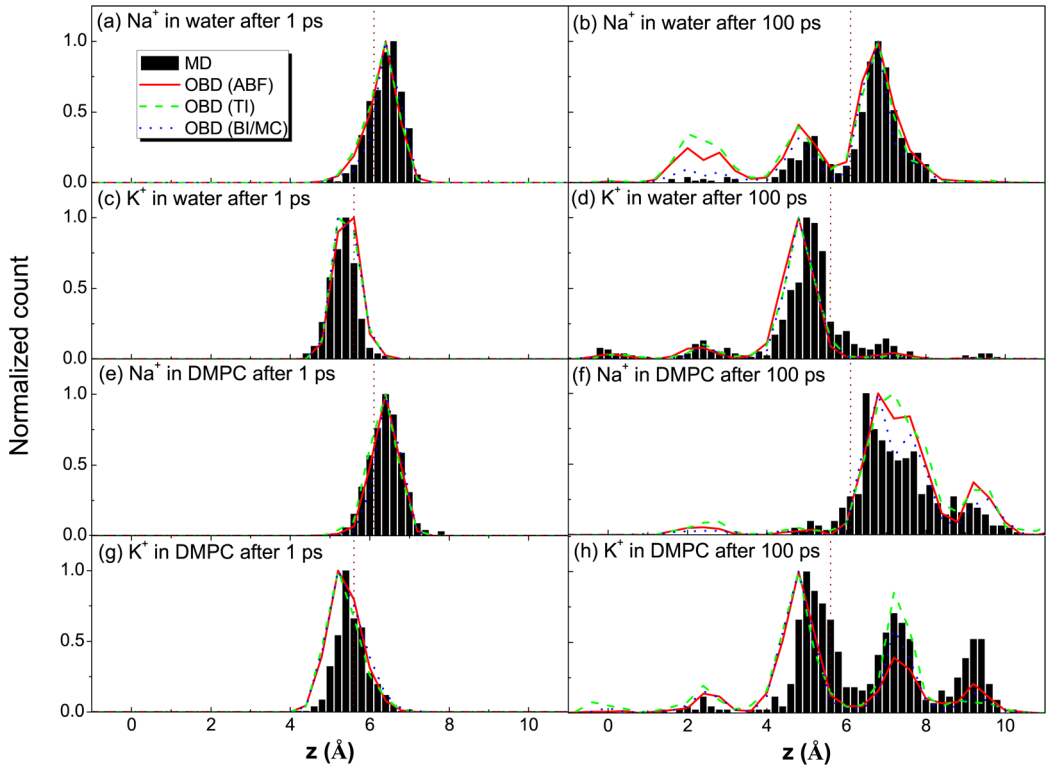


Figure 7. Distributions of the final z coordinates of Na^+ and K^+ after 1 and 100 ps that were initially located at z = 6.1 Å for Na^+ and at z = 5.6 Å for K^+ inside the CPN in the four systems. The three different PMF profiles obtained from the ABF, TI, and BI/MC methods were employed for $\tilde{G}_\alpha(z)$ in OBD simulations. The vertical broken lines represent the initial z coordinates of the cations which are slightly displaced away from a barrier top in the PMF curves.

303 K (the experimental temperature) remain unaltered at 298 K. Table 2 shows that the calculated conductances are nearly 4 or 5 times larger than the measured ones. One reason for the discrepancy can be attributed to the radius R_0 used for the effective cross-sectional area because the effective cross-sectional area inside the channel can be smaller than πR_0^2. To obtain a more realistic effective cross-sectional area, we performed 40 ns MD simulations for Na^+ and K^+ inside the CPN in the DMPC bilayers and calculated $\langle r_s^2 \rangle$, where r_s is defined as $r_s = r_i + r_a$. Here r_i is the intrinsic cation radius and r_a is the radial distance between the cation and the channel axis which is calculated as $r_a = \sqrt{x^2 + y^2}$, where x and y are relative coordinates from the channel axis. The Pauling radius was adopted for r_i which is 0.95 Å for Na^+ and 1.33 Å for K^+,[58] and r_a was directly evaluated from the MD simulations. The acquired mean values of $\langle r_s^2 \rangle$ for Na^+ and K^+ were 5.36 and 5.85 $Å^2$, respectively, which are smaller than 6.25 (=2.5^2) $Å^2$ used above. Thus, if we take those values as the effective radii, we will obtain conductances of 233 and 266 pS for Na^+ and K^+, respectively, slightly smaller than in Table 2.

Another issue of importance in this comparison is that the CPN is observed (in both experiments and MD simulations) to be tilted in the hydrated lipid bilayers.[2,9,14,59] As the channel axis of the CPN gets tilted, the area of the channel entrance becomes occluded in part by the membrane interface, and consequently, the ion currents will be reduced. Consideration of the tilt of the CPN, which is currently not taken into account in our study, will also reduce the conductance in the calculations. The periodic boundary conditions and the system size can also affect the disagreements between the calculated and measured conductances.[35] Although the individual calculated conductances of Na^+ and K^+ highly overestimate the experimental conductances, the ratio of the calculated conductances between K^+ and Na^+, 1.05, is close to the ratio of the experimental conductances, 1.18. Compared to the ratio of the experimental diffusion coefficients in the bulk between the two ions, 2.03/1.33 ≈ 1.53, this demonstrates that the simple bulk diffusion coefficients of ions alone cannot explain the difference in the conductance of ions and that the energetics as well as the local diffusivity of ions should be taken into account simultaneously.

To inspect the validity of the PMF and position-dependent diffusion coefficients of Na^+ and K^+ through the CPN in water and in a hydrated DMPC that we obtained, we performed overdamped Brownian dynamics simulations that require the PMF and diffusivity profiles, and compared the predictions from OBD simulations with the results from MD simulations. To do so with OBD and MD simulations, we initially positioned a single Na^+ at z = 6.1 Å and a single K^+ at z = 5.6 Å inside the CPN where the initial position for each cation is slightly displaced away from a barrier top in the PMF curves. Then, we ran a series of independent simulations, and collected the z coordinates of the cation after 1 and 100 ps from both simulation methods. In the OBD algorithm, the equation of motion for the cation α is given as

$$\dot{z}_\alpha(t) = -\frac{\tilde{D}_\alpha(z)}{k_B T} \cdot \frac{d\tilde{G}_\alpha(z)}{dz} + \frac{d\tilde{D}_\alpha(z)}{dz} + \zeta_\alpha(z, t) \quad (17)$$

where $\zeta_\alpha(z, t)$ is a Gaussian random noise with $\langle \zeta_\alpha(z, t)\zeta_\alpha(z, 0) \rangle = 2\tilde{D}_\alpha(z)\delta(t)$.[60] In eq 17, $\tilde{G}_\alpha(z)$ was obtained by fitting to the PMF profiles for the cations to a function given by

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$$\tilde{G}_\alpha(z) = \sum_{m=0}^{N} \{A_m \cos[2m\pi z/(z_f - z_i)] + B_m \sin[2m\pi z/(z_f - z_i)]\} \quad (18)$$

where A_m and B_m are fitting parameters and $z_i = -30.0$ Å and $z_f = 30.0$ Å. For $\tilde{G}_\alpha(z)$ in eq 17, the three different PMF profiles obtained from the ABF, TI, and BI/MC methods, respectively, were employed. The fitting function $D^{fit}(z)$ in eq 9 was adopted for $\tilde{D}_\alpha(z)$ in eq 17. For OBD simulations, a total of 1000 independent trajectories starting from an initial coordinate of $z = 6.1$ Å for Na^+ and $z = 5.6$ Å for K^+ were produced and the final z coordinates of the cations were collected for the calculation of the distributions. In MD simulations, a total of 500 independent trajectories for each system with the same initial z coordinate of the cation as in the OBD simulations were generated and the distributions of the final z coordinates of the cation were calculated. Figure 7 represents the distributions of the final z coordinates of Na^+ and K^+ after 1 and 100 ps obtained from the MD and OBD simulations. Compared with the MD simulation results, the OBD simulations predict the final distribution of Na^+ through the CPN both in water and in the DMPC bilayers after 100 ps quite well. In the case of K^+, the OBD results for the cation inside the CPN in water show good agreement with the MD simulation ones, while the OBD predictions for the cation inside the CPN in the DMPC are only in qualitative agreement with the MD results. It appears that the depth of the energy well located near 5.0 Å in the PMF curves of K^+ through the CPN in the DMPC bilayers is slightly overestimated and the depths of the energy wells near 7.2 and 9.2 Å are slightly underestimated, and as a result, the OBP simulations predict smaller populations near 7.2 and 9.2 Å than the MD simulations do. It is observed in both the MD and OBD simulations that the cations still stay around the initial position after 1 ps due to slow diffusion processes inside the channel, but after 100 ps, they are broadly distributed over several locations corresponding to the energy minima. Since the PMF profiles for the cations represent deeper energy wells in the middle of the channel in water than in the DMPC bilayers, the distributions of the cation positions after 100 ps are more shifted into the middle of the channel in water than in the bilayers. We also note that, although the initial positions of the cations in the four systems are very similar, the final distributions for the cations are quite diverse, demonstrating that small differences in the PMF and diffusivity profiles of ions as well as in the initial conditions such as the initial position can have significant impact on the dynamics and time evolution of ions through ion channels.

CONCLUDING REMARKS

In this work, the PMF profiles and position-dependent diffusion coefficients associated with transport of Na^+ and K^+ through a CPN in water and in hydrated DMPC bilayers were calculated using molecular dynamics simulations combined with the ABF method and BI/MC algorithm. In addition, the TI method was used for decomposition of the PMF profiles of the ions, and the dynamic friction kernel analysis was employed to investigate the contribution of each component in the systems to the position-dependent diffusion coefficients.

The PMF profiles and PMF decomposition profiles for the cations demonstrate that favorable interactions of the cations with both the CPN and lipid bilayers and dehydration free energy penalties are two competing factors that determine the free energy surface for ion transport through the CPN in water and in DMPC bilayers. The PMF decomposition profiles also reveal that the selectivity of the CPN to monovalent cations such as Na^+ and K^+ in the lipid bilayers is due to favorable interactions of the ions with CPNs and lipid bilayers that exceed dehydration penalties. It was observed in the PMF decomposition profiles that the favorable interactions and the dehydration penalties are most enhanced at the channel entrances due to interactions of the cations with the dangling carbonyl oxygens belonging to the outermost cyclic peptide rings.

Calculations of the position-dependent diffusion coefficients for the cations based on the BI approach show that the diffusion coefficients begin to decrease more than 10 Å away from the channel entrance and reach their minimum values near the channel entrances. The dynamic friction kernel calculations reveal that the highest contribution to the total dynamic friction kernels comes from the water−water dynamic friction kernel, demonstrating the strong effect of the channel water molecules on the diminished diffusion coefficients of the cations through the channel. The calculations also indicate that the decrease of the diffusion coefficients far outside the channel and the reduced coefficients inside the channel can be attributed to the dehydration process, to molecular rearrangements outside the channel, and to the coupling of the ion motions with the chain-structured water movements inside the channel.

Overestimation of the conductances of Na^+ and K^+ as determined from the PMF and position-dependent diffusion coefficient profiles for the ions implies that more accurate evaluations of the effective cross-sectional area and considerations of the tilt behavior of CPNs in lipid bilayers should improve the agreement of ion conductances with experiment. Despite the overestimation of the ion conductances, the conductance calculations and the comparisons between the OBD predictions and MD simulation results emphasize that the energetics and the local diffusivity of ions are two important factors that govern the ion dynamics, and that both are required to explain ion conductances through the CPN and other ion channels.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.6b09638.

One figure (Figure S1) to show the position-dependent diffusion coefficients of Na^+ and K^+ through the CPN in water and DMPC bilayers with different observation times Δt (PDF)

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Notes

The authors declare no competing financial interest.

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